MACHINE LEARNING · Responsible AI

Bias Detection Tool

Problem Statement

AI hiring models trained on historical data inherit and amplify human prejudice — systematically rejecting qualified candidates based on protected attributes like gender, race, and age.

Core Question: Does a hiring AI treat Male and Female candidates equally — or does it systematically favor one group?

2,000
Synthetic Candidates

+12%
Injected Gender Bias

3
Fairness Metrics

2
Mitigation Methods

Sonu Kumar & Aditya · April 2025

02 Motivation

Why Does This Matter?

Legal Risk

Title VII (US), GDPR Art. 22 (EU), Equality Act (UK) — biased AI tools can trigger regulatory action & lawsuits.

Talent Loss

Systematic rejection of qualified candidates from underrepresented groups — economically damaging and ethically wrong.

Feedback Loops

Biased training data → biased model → biased hiring → even more biased future data. Amplification over time.

Reputational Damage

afte

Real Case: Amazon AI Recruiter

- Amazon built an AI tool to automate CV screening for engineering roles.
- The model was trained on 10 years of historical hiring data.
- Since most past hires were male, the model learned to penalise female applicants.
- It downgraded CVs with the word women's — e.g. women's chess club.
- Amazon scrapped the tool entirely in 2018 after the bias was discovered.
- This project builds the detection & mitigation pipeline that could have caught this.

15–25%
Higher revenue for gender-diverse companies (McKinsey)

2018
Year Amazon scrapped its biased AI recruiter

Art. 22
GDPR right to explanation for automated decisions

03 Dataset & Features

AI-Powered Job Market Insights Dataset (Synthetic)

Feature	Description	Role
Age	Candidate age (22–55)	Input
Gender	Male / Female	⚠️ Sensitive
EducationLevel	High School → PhD	Input
ExperienceYears	Years of work exp.	Input
InterviewScore	Performance score (0–100)	Input
SkillScore	Technical assessment (0–100)	Input
PersonalityScore	Personality score (0–100)	Input
DistanceFromCompany	Commute distance (km)	Input
HiringDecision	1=Hired, 0=Not Hired	🎯 Target

How Bias Was Injected

Intentional gender bias mirrors real-world patterns:

Base hire probability computed from:

- InterviewScore × 0.003
- SkillScore × 0.002
- ExperienceYears × 0.015
- EducationLevel × 0.05
- DistanceFromCompany × −0.003

Gender boost applied:

- Male candidates → +0.12 added
- Female candidates → +0.00

n = 2,000 candidates · 75% train · 25% test

Result: Male hire rate ~12% higher than female even with equal qualifications

04 Approach & Pipeline

1

Data Generation

Synthetic 2000-candidate hiring dataset with controlled +12% male bias injected via NumPy probability engine

2

Preprocessing

Ordinal encoding (education), binary encoding (gender), one-hot (strategy), StandardScaler normalization

3

EDA & Bias Analysis

Visualize hiring rates by gender, correlation heatmap, interview/skill score distributions per group

4

Baseline Model

Logistic Regression (lbfgs, max_iter=1000) trained on scaled features — establishes the 'before' picture

5

Fairness Evaluation

Fairlearn MetricFrame — Demographic Parity, Disparate Impact Ra

05 Implementation Details

Tech Stack

Python 3.8+ — Core language

Jupyter Notebook — Dev env

Pandas — Data manipulation

NumPy — Numerical computing

Scikit-learn — ML models

Fairlearn — Fairness metrics

SHAP — Explainability

Matplotlib / Seaborn — Viz

Fairness Metrics

Demographic Parity Diff → target: 0

Demographic Parity Ratio → target: 1

Equalized Odds Diff → target: 0

4/5 Rule (80% Rule) — legal standard

Group Accuracy per gender

Group Recall (TPR) per gender

Group FPR per gender

Fairlearn MetricFrame summary

Mitigation Techniques

Method 1: Sample Reweighting

- Pre-processing technique
- Expected size = n / (groups × labels)
- Weight = expected / actual size

Method 2: ExponentiatedGradient

- In-processing technique
- DemographicParity constraint
- eps = 0.01 (strict fairness)

SHAP LinearExplainer used for model interpretability · 200 test samples subsampled for SHAP computation speed · Random State = 42

06 Results — Bias Detection

71%
Baseline Accuracy
Overall model performance

0.23
DP Difference
Target = 0 · BIASED ⚠️

0.71
DP Ratio
Below 0.80 · 4/5 Rule VIOLATED

0.19
Eq. Odds Diff
Target = 0 · BIASED ⚠️

Group-Level Performance — Baseline Model

Group	N (test)	Selection Rate	Accuracy	Recall (TPR)	FPR
Male	277	68%	73%	71%	34%
Female	223	45%	69%	52%	19%
Gap (M−F)	—	**+23%** ⚠️	+4%	**+19%** ⚠️	+15%

🔍 SHAP Insights

InterviewScore → Highest impact (legitimate)

SkillScore → 2nd highest (legitimate)

ExperienceYears → 3rd (legitimate)

Gender → Significant SHAP value ❌

Blue (Female) dots → Left (−SHAP)

Red (Male) dots → Right (+SHAP)

07 Results — After Mitigation

Before vs. After Bias Mitigation

Metric	Baseline	Reweighted	ExpGradient	Ideal
Accuracy	71%	69%	68%	—
DP Difference ↓	**0.23** ❌	0.14 ⚠️	**0.03** ✅	0
DP Ratio ↑	**0.71** ❌	**0.82** ✅	**0.96** ✅	1.0
Eq. Odds Diff ↓	**0.19** ❌	0.11 ⚠️	**0.04** ✅	0
4/5 Rule	**VIOLATED** ❌	**PASSED** ✅	**PASSED** ✅	PASS

−39%

Sample Reweighting

Reduction in DP Difference with only 2% accuracy drop

−87%

ExponentiatedGradient

Reduction in DP Difference with only 3% accuracy drop

4/5 ✅

Both Methods

Legal 4/5 Rule satisfied after mitigation

08 Main Challenges

01 Accuracy–Fairness Trade-off

Enforcing strict fairness constraints (eps=0.01) always reduces accuracy. Finding the right eps value that satisfies legal standards (≥80% DP Ratio) while maintaining useful model performance is a key tension in all fairness work.

02 Incompatible Fairness Criteria

Demographic Parity, Equalized Odds, and Equal Opportunity are mathematically proven to be mutually incompatible when base rates differ between groups. A model cannot satisfy all three simultaneously — requiring deliberate prioritisation.

03 Proxy Variables & Indirect Bias

Even after removing Gender from features, the model can still discriminate through proxy variables correlated with gender (e.g., career gaps, certain universities). Detecting indirect bias requires extensive correlation analysis.

04 Synthetic vs Real Data Limitations

The dataset is synthetic, meaning the bias is deliberately controlled. In real organizational data, bias is hidden, complex, and multi-dimensional — making detection much harder and requiring domain expertise to interpret results.

05 SHAP Computational Cost

Computing SHAP values for all candidates is expensive. Even with LinearExplainer (efficient for Logistic Regression), we needed to subsample 200 test candidates. More complex models like Random Forest would be significantly slower.

06 Intersectional Bias

Our tool focuses on single-attribute bias (Gender). Real-world bias is intersectional — a candidate who is both female AND from a minority ethnicity AND over 45 may face compounded disadvantage that single-axis analysis cannot capture.

09 Future Scope

Short Term

- Test on real-world organizational hiring datasets
- Extend to race, age, and intersectional bias detection
- Add confidence intervals to all fairness metrics

Medium Term

- Build an interactive HR dashboard for upload & audit
- Explore post-processing: per-group threshold adjustment
- Add adversarial debiasing as a 3rd mitigation method

Long Term

- Continuous monitoring system for deployed models
- Expand beyond hiring: credit, healthcare, criminal justice
- Contribute to open-source AI fairness standards

💡 Vision

Transform this notebook into a plug-and-play fairness audit tool that any HR team can use — upload a model's predictions, get a full fairness report, flagged violations, and recommended mitigations — with zero ML expertise required.

Key Takeaways

- ✅ The baseline model inherited & amplified the +12% gender bias present in training data
- ✅ Fairlearn MetricFrame clearly quantified disparity — DP Difference 0.23, 4/5 Rule violated
- ✅ Sample Reweighting reduced bias by 39% with only a 2% accuracy loss
- ✅ ExponentiatedGradient reduced bias by 87% — best fairness, marginal accuracy cost
- ✅ SHAP revealed Gender contributing comparable weight to ExperienceYears — direct evidence
- ✅ Fairness & accuracy are not mutually exclusive — mitigation makes AI systems both fair AND useful

87%
Max bias reduction

3%
Accuracy trade-off

0.96
Final DP Ratio

✅
4/5 Rule satisfied